

AROON KUMAR

Artificial Intelligence Engineer

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SUMMARY

AI/ML Engineer with **1+ years of experience** building intelligent systems across **machine learning, computer vision, and LLM applications**. Experienced in developing scalable AI pipelines and production-ready ML services using **Python, PyTorch, TensorFlow, and OpenCV**. Built an **AI-powered PCB visual inspection system** achieving **91.2% F1-score** for multi-label defect detection using transfer learning (ResNet-18). Hands-on experience with **RAG architectures, speech AI pipelines, and data-driven ML models**, along with API-based deployment and microservices. Passionate about applying **computer vision and machine learning** to solve real-world industrial problems and building scalable AI systems.

EXPERIENCE

Database Operations

SZABIST Islamabad 02/2026 - Present Islamabad, Pakistan

A university department managing centralized student databases and academic records.

- Manage and maintain **student databases containing 25,000+ records** for current and alumni students.
- Execute **database queries, validation, and updates** to ensure data integrity.
- Work with **API-based integration to the centralized university server in Karachi** for secure data synchronization.
- Support administrative systems by retrieving and processing **large-scale datasets efficiently**.

AI/ML Solutions Executive (Intern → Part-time)

Prudential Solutions 08/2025 - 02/2026 Islamabad, Pakistan

AI and ML solutions company building intelligent systems and automation platforms.

- Built and deployed **voice chatbots, speech-to-speech AI systems, and LLM applications**.
- Led development of ITP ConvertX multi-city chatbot supporting 5 cities and 3 languages.
- Designed **RAG pipelines for document ingestion and dynamic knowledge retrieval**.
- Implemented **multi-model AI architecture (GPT + Mistral fallback)** for reliability.
- Built **Redis-powered API Gateway**, reducing latency 50% for 200+ users.
- Automated **CI/CD pipelines**, reducing manual work by 70%.

Data Science Intern

10Pearls 12/2025 - 02/2026 Islamabad, Pakistan

Global software company specializing in AI and machine learning solutions.

- Developed **end-to-end AQI forecasting system** using ML models.
- Built **automated ML pipelines** for feature engineering, training, inference.
- Implemented **CI/CD workflows** for daily model retraining and deployment.
- Achieved **MAE <10% and R² >0.85** in predictions.
- Interactive **Streamlit dashboard** with **Explainable AI (SHAP, LIME)**.

SKILLS

Computer Vision	Deep Learning
Machine Learning	Python C++
PyTorch	OpenCV TensorFlow
Scikit-learn	Transfer Learning
Image Processing	Data Augmentation
Model Evaluation	FastAPI REST APIs
Docker	Pandas Matplotlib

PROJECTS

AI-Powered PCB Visual Inspection System

01/2026 - 03/2026 Islamabad, Pakistan

<https://github.com/AroonKumarr/ai-pcb-quality-inspection>

This side-project is an AI-powered PCB inspection system that detects multiple defects on circuit boards automatically.

- Built with PyTorch and OpenCV, the system achieved 91.2% F1-score, enabling fast and reliable industrial quality checks.

Air Quality Index Forecasting System

12/2025 - Present Islamabad, Pakistan

<https://air-quality-ml-mlops.streamlit.app/>

This side-project is an ML-powered AQI forecasting system that predicts air quality for the next three days.

- Built with Python, Scikit-learn, and automated pipelines, the system achieved MAE <10% and R² >0.85 for forecasts.

EDUCATION

Bachelor in Computer Science

SZABIST Islamabad

02/2022 - 02/2026 Islamabad, Pakistan